Arijit Ghosh

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Education

PhD in Computer Vision — Since Dec 2024

École Nationale des Ponts et chaussées, IP Paris and École Polytechnique, IP Paris

Supervisors: Prof. Dr. David Picard and Prof. Dr. Vicky Kalogeiton
Thesis: Understanding generative models for visual tasks.

M.Sc. in Artificial Intelligence — Oct 2021 - Aug 2024

Friedrich Alexander Universitat Erlangen–Nürnberg

Final Grade: **1.3** (Best: 1.0, Worst: 4.0)
Minor: Artificial Intelligence in Biomedical Engineering
Master Thesis: Exploring Disentagled Structure-Texture Generative Models on Image-To-Image Translation Task.

B. Tech. in Electronics and Communication Engineering — Aug 2017 - Jul 2021

Maulana Abul Kalam University of Technology (formerly West Bengal University of Technology)

Final Grade: **9.24** (Best: 10.0, Worst: 5.0)

Research Experience

Research Assistant — Jun 2024 - Nov 2024

École Nationale des Ponts et chaussées, IP Paris (Supervisor: **Prof. Dr. David Picard**)

Student Teaching and Research Assistant — Jan 2022 - May 2024

IDEA Lab, FAU, Erlangen–Nürnberg (Supervisor: **Prof. Dr. Bernhard Kainz**)

Student Research Assistant — Mar 2023 - May 2024

Fraunhofer IIS, Nürnberg (Supervisor: **Mark Deutel, M.Sc.**)

Work Experience

Working Student - Deep Learning and Computer Vision — Sep 2022 - Feb 2023

Primetals Technologies, Erlangen (Supervisor: **Dominik Wassermann, M.Sc.**)

Mentor - Generative Adversarial Networks Specialization — Aug 2021 - Dec 2021

Deeplearning.AI

Publications

Arijit Ghosh*, Lucas Degeorge*, Paul Couairon, Alexei A. Efros, Vicky Kalogeiton and David Picard. **"Dino Forcing Flow Models: Do not denoise what you can predict"**. *arXiV, 2026*.

Lucas Degeorge*, Paul Couairon*, **Arijit Ghosh***, Alexei A. Efros, David Picard and Vicky Kalogeiton. **"Balancing Frequencies and Pixels in Flow Matching"**. *NeurIPS, 2026*.

Nicolas Dufour, Lucas Degeorge*, **Arijit Ghosh***, Vicky Kalogeiton and David Picard. **"MIRO: MultI-Reward cOnditioned pretraining improves T2I quality and efficiency"**. *ICML, 2026*.

David Picard, Nicolas Dufour, Lucas Degeorge, **Arijit Ghosh**, Davide Allegro, Tom Ravaud, Yohann Perron, Corentin Sautier, Zeynep Sonat Baltaci, Fei Meng, Syrine Kalleli, Marta López-Rauhut, Thibaut Loiseau, Ségolène Albouy, Raphael Baena, Elliot Vincent, Loic Landrieu . **"PoM: A Linear-Time Replacement for Attention with the Polynomial Mixer"**. *Findings Track, CVPR 2026*.

Lucas Degeorge*, **Arijit Ghosh***, Nicolas Dufour, David Picard, and Vicky Kalogeiton. **"How far can we go with ImageNet for Text-to-Image generation?"**. *arXiv, 2025*.

Technical and Specialized Skills

Languages/Frameworks: Confident with Pytorch, Pytorch-Lightning, Monai, Python, Git.

Fields of Interest: Conditional Image Generation, Multimodal generative models, Data efficient generative models, Unpaired condition to image generation

Projects

Master's Project-1 : Segmentation with a punch via VoxelMorph and Registration.
Supervisor: **Prof. Dr. Bernhard Kainz**

Master's Project-2 : Fake it till you Can't make it.
Supervisor: **Prof. Dr. Bernhard Egger**

Achievements

Winner of Researchathon 2021 in Healthcare Track **July 2021**

Referees

Prof. Dr. David Picard : david.picard@enpc.fr
Prof. Dr. Vicky Kalogeiton : vicky.kalogeiton@polytechnique.edu
Prof. Dr. Alexei A. Efros : efros@eecs.berkeley.edu
Prof. Dr. Bernhard Kainz : bernhard.kainz@fau.de
Prof. Dr. Bernhard Egger : bernhard.egger@fau.de
Mr. Mark Deutel, M.Sc.: mark.deutel@iis-extern.fraunhofer.de